

Existence Theorems for Inclusions

Condition 1: Compactness F is compact domain Ω or compact feasible set

Condition 2: Convexity $F(\omega)$ is convex for all ω

Condition 3: Measurability F has measurable selections

Condition 4: Growth condition $F(\omega)$ with g integrable

Main Theorem (Existence Result)

If conditions 1-4 hold, then the inclusion

$$\frac{dx}{dt} \in F(t, x(t))$$

has at least one solution

Applications:

- Optimal control problems with constraints
- Delay differential equations
- Nonlinear partial differential equations
- Stochastic systems with uncertainties